Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti |
Virginia, USA (commutable to DC)

PROFESSIONAL SUMMARY

Quantitative researcher and data analyst with 8 years working inside the systems that Urban Institute's Housing and Communities Division studies - HUD income-limit imports, LIHTC eligibility calculations, AMI-based rent determinations, and affordable housing compliance workflows for 100K+ housing units at Entrata. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026), with hands-on experience building reproducible Python and R data pipelines, statistical models ($R^2 = 0.977$), and Tableau/ggplot2/Quarto visualizations across large administrative and survey datasets. Comfortable using AI tools responsibly to improve coding efficiency while protecting data confidentiality. Strong interest in housing policy, community economic development, and structural inequality - informed by direct operational experience with affordable housing programs.

CORE COMPETENCIES

Housing Policy · HUD · LIHTC · AMI · Affordable Housing | Quantitative Research · Statistical Modeling | Python · R · SQL · Reproducible Code | Large Administrative & Survey Datasets | Data Visualization · Tableau · ggplot2 · Quarto | Data Cleaning · Quality Assurance · Documentation | AI Tools · Responsible Use · Workflow Efficiency

WORK EXPERIENCE

George Mason University - Research

Aug 2024 - Aug 2025

Research Associate

- **Built reproducible multi-source prediction models** (Python, R) integrating WHO Global Health Observatory and UK Road Safety administrative datasets across 190+ countries - $R^2 = 0.977$ regression accuracy; designed the full analytic pipeline from data collection, cleaning, and merging through hypothesis testing, model validation, and interpretation of results for non-technical audiences.
- **Applied Isolation Forest anomaly detection** to automate data integrity checks across 3 research pipelines, surfacing inconsistencies and data limitations that manual review missed - improving downstream model reliability and quality assurance across all analytic outputs.
- **Produced tables, charts, and dashboards** for faculty leadership using Tableau, ggplot2, and Quarto - presenting descriptive statistics and key patterns clearly; improved research comprehension by 40% across two cohorts and non-technical stakeholder audiences.
- Contributed to literature reviews and research summaries; documented analytic decisions, reconciled results, and flagged data limitations across all research pipelines; collaborated with research teams to prioritize and complete multiple concurrent projects.

George Mason University - College of Engineering

Aug 2025 - May 2026 · Part-time

Graduate Teaching Assistant

- Delivered technical instruction to 600+ students across multiple sections - communicating data analysis, Python programming, database design, and statistical concepts clearly to varied skill levels; provided code review feedback aligned with course standards and rubric documentation.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 - Aug 2024 · Remote

Senior Software Engineer - Affordable Housing & Compliance Data

- **Owned HUD income-limit data pipelines for 7 years** - transformed annual federal source datasets (HUD income limits, tax credit thresholds, AMI tables) into ~180,000 domain records powering eligibility, rent, HAP, and LIHTC calculations for 100K+ housing units. Every release required baseline validation, reconciliation against prior-year data, and quality assurance before production. Zero compliance violations across the ownership period.
- **Managed and analyzed large administrative datasets** (1M+ property records, multi-tenant schemas) using Python, SQL, and PL/SQL - optimized 200+ queries (30% performance gain) through execution plan analysis, indexing, and stored procedure refactoring; documented all analytic decisions and schema changes.
- **Built automated data quality checks and audit pipelines** in Python and PL/SQL - eliminated manual reconciliation for gross-rent updates across thousands of tenant records; idempotency guarantees prevented double-processing errors across annual rent-schedule changes.
- Led AWS S3 cloud migration for housing document records (100% PHI compliance, zero security incidents); built 56 state-specific certification document templates mapping to housing program requirements across LIHTC, HCV, and Section 8 workflows.
- Mentored 5+ engineers; conducted 300+ code reviews; built internal debugging and validation tools used across the compliance data team.

Entrata India (Db Xento Systems)

Jan 2017 - Sep 2018 · Pune

Associate / Trainee Software Engineer

- Built affordable housing data modules (income limits, 1.6M HUD/Tax Credit records across 5 years), document management, and compliance reporting workflows; redesigned document storage database architecture for 180K+ records.

SELECTED PROJECTS

CrashClarity: Large-Scale Road Safety Statistical Analysis (104K Records)

2025 | Python, R, scikit-learn, Random Forest, Logistic Regression, GitHub Pages

venkatalakshmiparimala.github.io/Portfolio

End-to-end quantitative analysis of 104K+ UK road accident administrative records - built a Random Forest + Logistic Regression model ensemble with feature engineering across weather, road condition, and temporal variables (85%+ accuracy); produced tables, descriptive statistics, and an interactive data visualization dashboard. Demonstrates the full research pipeline: raw data ingestion, cleaning, merging, modeling, quality assurance, and clear presentation of findings for non-technical audiences.

Leasel: Affordable Housing Document Intelligence Platform

2026 | Python, FastAPI, TF-IDF, OCR, SQLite, HUD API

github.com/venkatalakshmiparimala/Lease_complaince_rag

Built a document analysis platform for affordable housing lease and compliance PDFs (1,240 residential leases across 5 US states + federal regulations) - OCR-based ingestion, section-aware chunking, TF-IDF retrieval, and automated clause extraction. Connected to the live HUD income-limit API for real-time regulatory data. Designed with responsible AI use: AI extracts and reasons; human or deterministic logic makes every final eligibility and compliance call. Results: 93.6% precision, 100% citation correctness, 0% unsupported-claim rate.

AgentProbe: AI-Assisted Analytic Workflow Toolkit Open Source

2025 | Python, FastAPI, CI/CD (GitHub Actions)

github.com/venkatalakshmiparimala/AgentProbe

Built an open-source toolkit for evaluating and quality-assuring AI-generated outputs - 5-metric eval engine (correctness, groundedness, hallucination rate), CI-integrated regression gate, and multi-provider LLM routing. Demonstrates responsible AI tool use: every AI-generated output is verified against ground truth before being trusted in downstream analysis.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026

GPA: 3.87 | Coursework: Statistical Modeling, Machine Learning, Data Visualization, Database Systems, Geospatial Analysis, Big Data Engineering

B.Tech Computer Science & Engineering - Jawaharlal Nehru Technological University, Anantapur — 2017

SKILLS

Statistical & Analytic: Python, R, SQL, PL/SQL, statistical modeling, regression analysis, descriptive statistics, Isolation Forest, Random Forest, Logistic Regression, data cleaning, data merging, quality assurance, reproducible code, longitudinal data, administrative datasets, survey data

Data Visualization: Tableau, Power BI, ggplot2, Quarto, Matplotlib, Seaborn, Plotly - tables, charts, maps, dashboards for technical and non-technical audiences

Housing Domain: HUD income limits, LIHTC, AMI-based rent calculation, Section 8/HCV, affordable housing eligibility, HAP, housing compliance workflows, urban planning data, federal housing datasets

Tools & Practices: Git, GitHub Actions (CI/CD), Docker, AWS S3, Jupyter, Excel, responsible AI tool use, Agile/Scrum, documentation, code review